

Quaternions

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1 Theoretical introduction

A quaternion q is essentially a number such as $q \in \mathbb{H}$, where:

$$\mathbb{H} := \{w + xi + yj + zk : w, x, y, z \in \mathbb{R}, i^2 = j^2 = k^2 = ijk = -1\}$$

2 Hamilton's equations

3 Quaternions arithmetic

As a quaternion is nothing but a number, operations among quaternions of course are allowed. For a quaternion $q_0 = a_0 + b_0i + c_0j + d_0k$ and another quaternion $q_1 = a_1 + b_1i + c_1j + d_1k$:

$$\begin{aligned}\lambda q &= \lambda(w + xi + yj + zk) = \lambda w + \lambda xi + \lambda yj + \lambda zk, \forall \lambda \in \mathbb{F}, q \in \mathbb{H} \\ q_0 + q_1 &= (w_0 + x_0i + y_0j + z_0k) + (w_1 + x_1i + y_1j + z_1k) \\ &= w_0 + w_1 + (x_0 + x_1)i + (y_0 + y_1)j + (z_0 + z_1)k \\ q_0 - q_1 &= q_0 + (-q_1) = (w_0 + x_0i + y_0j + z_0k) + (-w_1 - x_1i - y_1j - z_1k) \\ &= w_0 - w_1 + (x_0 - x_1)i + (y_0 - y_1)j + (z_0 - z_1)k \\ q_0 q_1 &= (w_0 + x_0i + y_0j + z_0k)(w_1 + x_1i + y_1j + z_1k) \\ &= w_0 w_1 + w_0 x_1 i + w_0 y_1 j + w_0 z_1 k + w_1 x_0 i + x_0 i x_1 i + x_0 i y_1 j + x_0 i z_1 k \\ &\quad + w_1 y_0 j + y_0 j x_1 i + y_0 j y_1 j + y_0 j z_1 k + w_1 z_0 k + z_0 k x_1 i + z_0 k y_1 j + z_0 k z_1 k \\ &= w_0 w_1 + w_0 x_1 i + w_0 y_1 j + w_0 z_1 k + w_1 x_0 i - x_0 x_1 + x_0 y_1 k - x_0 z_1 j + w_1 y_0 j \\ &\quad - y_0 x_1 k - y_0 y_1 + y_0 z_1 i + w_1 z_0 k + z_0 x_1 j - z_0 y_1 i - z_0 z_1 \\ &= w_0 w_1 - x_0 x_1 - y_0 y_1 - z_0 z_1 + i(w_0 x_1 + w_1 x_0 + y_0 z_1 - z_0 y_1) \\ &\quad + j(w_0 y_1 - x_0 z_1 + w_1 y_0 + z_0 x_1) + k(w_0 z_1 + x_0 y_1 - y_0 x_1 + w_1 z_0)\end{aligned}$$

4 Magnitude

Just like in complex numbers or vectors, quaternions also have a norm or magnitude that is computed right in the same way or using the same logic as them. By using the pythagorean theorem:

$$\|q\|^2 = w^2 + x^2 + y^2 + z^2 \implies |q| = \sqrt{w^2 + x^2 + y^2 + z^2}$$

5 Conjugate of a quaternion

The conjugate of a quaternion q is the same concept as the conjugate of a complex number, we just flip the symbol from the imaginary components:

$$q^* = w - xi - yj - zk$$

There is an interesting property of the conjugate, if we multiply qq^* , we get:

$$\begin{aligned} qq^* &= (w + xi + yj + zk)(w - xi - yj - zk) \\ &= w^2 - wxi - wyj - wzk + wxi + x^2 - xyk + xzj + wyj + yxk + y^2 - yzi \\ &\quad + wzk - zxj + zyi + z^2 \\ &= w^2 + x^2 + y^2 + z^2 = |q|^2 \end{aligned}$$

By expanding all terms we see that the majority of them can be canceled out, returning simply the squared magnitude of the quaternion.

6 Inverse of a quaternion

The interesting part here is nothing but the inverse as the inverse is defined as another quaternion q^{-1} such as $q^{-1}q = 1$:

$$q^{-1}qq^* = q^* \implies q^{-1}|q|^2 = q^* \implies q^{-1} = \frac{q^*}{|q|^2}$$

$$\therefore q^{-1} = \frac{q^*}{|q|^2}$$

We see that if $|q|^2 = 1$, which implies that $|q| = 1$, $q^{-1} = q^*$.